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F25-BSAI-0054

Lab Assignment # 01

07-10-2026

Submitted To: Sir Usama

CamScanner

Date: ----------

Assignment # 01

Question no. 1:-

Code

```
Students = []
def result (p, ai, math):
    total = p + ai + math
    per = total/3

if per >= 80:
    grade = "A"
elif per >= 70:
    grade = "B"
elif per >= 60:
    grade = "C"
elif per >= 50:
    grade = "D"
else:
    grade = "F"
return total, per, grade

# Input 5 students

for i in range(5):
```

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```
name = input("Enter Name")
roll = input("Enter roll Number")
p = float(input("python marks"))
ai = float(input("AI marks"))
maths = float(input("Maths marks"))

total, per, grade = result(p, ai, math)

students.append({
"name": name, "roll": roll, "total": total,
"percentage": per, "grade": grade
})
```

Display results

```
for s in students:
    print(s)
```

Highest Percentage

```
top = max(students, key = lambda s:
          s["Percentage"])
print("Highest Percentage:")
print(top)
```

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save result

```
with open ("result .txt " , "w") as file:
        for s in students:
        file .write (str (s))
    file. write (" Highest Marks")
    file . write (str (top))
```

Question no. 2:-

Code

```
def analyze (s):
        w =s. split ( )
        u = set (w)
        v = sum (c. lower ( ) in for c in s)
print ( "Words:", len (w))
print ("characters :", len (s))
print ("longest :", max (w , key =len))
print ("shortest :", min (w, key =len))
print (" vowels:" , v )
print (" unique words:" , u)
print ("Alphabetical :", sorted (u))

with open ("word-analysis. txt"; "w") as f:
f. write (str ( [len(w), len (s), max(w, key=l
```

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```
    min (w, key=len), v, u, sorted (u) ]))
s = input ("Enter sentence:")
        analyze (s)
```

Question no. 3:-

Code

```
def bill (products):
    total = 0
for p in products:
p["total"] = p["price"] * p["quantity"]
    total += p["total"]

discount = total * 0.10 if total > 10000
            else 0
final = total - discount
expensive = max (products, key - lambda p:
                p ["price"])

print ("Total Bill", total)
print ("Discount ", discount)
print ("Final bill", final)
print ("Most Expensive", expensive ["name"])

with open ("bill.txt", "w") as f:
    f.write (str (products))
```

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```
f.write("Final Bill :" + str(Final))
products = []
for i in range(5):
    name = input("Product Name")
    price = float(input("price"))
    quantity = int(input("Quantity"))

    products.append({
                    "name": name,
                    "price": price,
                    "quantity": quantity
                    })
bill(products)
```

Question no. 4:-

Code

```
def add(a, b):
    return a+b
def sub(a,b):
    return a-b
def mul(a,b):
    return a*b
def mod(a,b):
    return a%b
```

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```
def div (a, b):
    return a/b

while True:
    a = input ("Enter 1st Number")
    b = input ("Enter 2nd Number")
    op = input ("Enter operation (+, -, *, /, %)")

    if op == "+":
        print (add (a, b))
    elif op == "-":
        print (sub (a, b))
    elif op == "*":
        print (mul (a, b))
    elif op == "/":
        print (div (a, b))
    elif op == "%":
        print (mod (a, b))
    else
        print ("Invalid operation")
    again = input ("Again (Yes/No): ")
    if again.lower () != "yes":
        break
```

Question no 5:.

Code

```
menu = { 1: ("Burger", 500), 2: ("pizza", 800)
         3: ("Fries", 250), 4: ("Drink", 150)}
def order ():
    total = 0
  while True:
print("1. Burger 2. Pizza 3. Fries 4. Drink 5. Exit")
    n = int (input ("choice :"))
  if n == 0:
    break
  elif n in menu:
      q = int (input ("Quantity"))
      total += menu[n][1] * q
  else:
    print ("Invalid choice")
    print ("Final Bill", total)

  with open ("Orders.txt", "w") as f:
  f.write ("Final Bill" + str (total))
    order ()
```

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